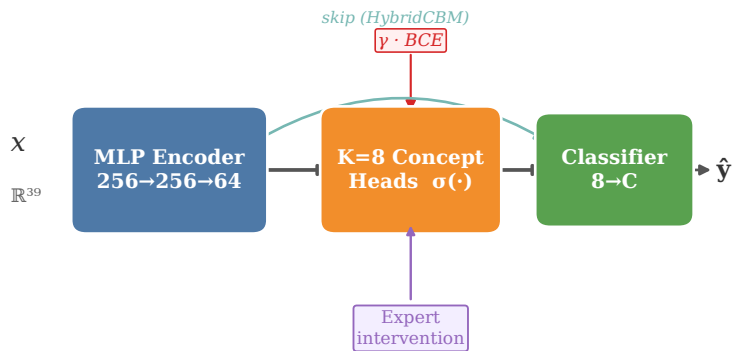


(a) Concept Bottleneck Models



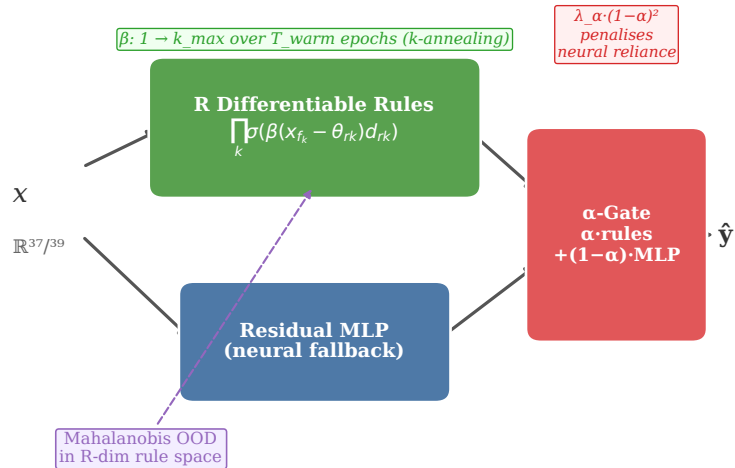
JointCBM encoder + concept + classifier end-to-end

SequentialCBM Concept heads frozen, classifier trained after
Test-time intervention on $\hat{c}$
OOD: Mahalanobis in 8-dim concept space

HybridCBM encoder skip appended to concept predictions

Post-hoc CBM frozen encoder + linear concept probes

(b) NeSy-NIDS



Thresholds θ_{rk} learned jointly | STE binarisation: 100% crisp rule activations at $k=k_{\max}$